



# On the accurate calculation of the dielectric constant from molecular dynamics simulations: The case of SPC/E and SWM4-DP water

Orsolya Gereben, László Pusztai \*

Research Institute for Solid State Physics and Optics, Hungarian Academy of Sciences, P.O.B 49, H-1525 Budapest, Hungary

## ARTICLE INFO

### Article history:

Received 23 November 2010

In final form 13 February 2011

Available online 17 March 2011

## ABSTRACT

The effect of the applied trajectory length on the convergence of the static dielectric constant was examined for the SPC/E water model with different system sizes. Very long simulation times of 6–8 ns were employed in order to track the convergence of this property. Temperature dependence and isotope effects were also investigated. A simulation for the polarizable SWM4-DP model was also carried out to compare the effect of different potential models. It is clearly shown that trajectories shorter than about 6 ns are not sufficient for a sufficiently accurate determination of the dielectric constant of these water models.

© 2011 Elsevier B.V. All rights reserved.

## 1. Introduction

Computer simulation of aqueous systems, including electrolyte and protein solutions, has been (and will increasingly be) an indispensable technique for the investigation of these systems. A key element of these computations is the water interaction potential model applied and therefore, precise knowledge concerning the behavior of the potentials, with respect to changing system size and simulation length, is of utmost importance.

Properties of the SPC/E [1–5] and SWM4-DP [6] water potential models have already been investigated thoroughly. The SPC/E model was chosen here, too, for its simplicity and popularity when studying aqueous systems, while the polarizable SWM4-DP model was selected as this model performed very well in our previous investigations regarding the structure of salt solutions [7,8]. As during these investigations we found, as side effects, quite substantial variances due to the application of different trajectory lengths while calculating the static dielectric constant, we decided to perform a new, systematic investigation on pure water. Our aim was not so much to give any further information on these specific models, but rather to exploit them as examples for studying the effects of different simulation parameters, mainly those of the applied trajectory length and system size during the determination of properties such as the static dielectric constant.

The dependence of the static dielectric constant on the system size using periodic boundary conditions, as well as reaction field, was previously investigated by Morrow [9]. He found that for a lattice of dipoles the static dielectric constant increases with increasing system size and slowly converges to the theoretical value of

infinite system size under periodic boundary conditions. For the same lattice system, using a reaction field with  $R_c$  cutoff distance, the convergence to the limit of infinite system size with increasing cutoff distance was not monotonous. In both cases large systems ( $\sim 10\,000$  dipoles for the periodic boundary condition model and  $R_c$  larger than half of the system size) could produce dielectric constants within 10% of the limiting value of infinite system size. For two small systems (125 and 343 molecules) Casulleras [10] found that the dielectric constant increases with increasing system size in molecular dynamics simulations performed for liquid methanol, using relatively short (500 ps) simulation lengths. It is fair to state that concerning the calculation of dielectric properties there are still quite a few ambiguities (see below) and thus, a systematic study by varying the system size and the length of the simulated trajectory (that is, the volume of configuration space explored) is rather timely.

## 2. Simulation details

The primary water potential in our investigations was the SPC/E model [1], at a temperature of 293 K. The simulations were carried out in the NVT ensemble under periodic boundary conditions with an atomic number density of  $\rho = 0.099\text{ Å}^{-3}$ . Four different system sizes over a wide range were used, containing 216, 518, 2000 and 16 000 water molecules; these systems are denoted by H1, H2, H3, H4, corresponding to the increasing number of molecules. To investigate the temperature dependence the H3-T model at 298 K was created, which calculation was otherwise identical to that of the H3 model. Simulation H1-O was performed with 216 water molecules at 298 K, used slightly different run parameters applying long-range dispersion correction for the energy (as implemented in the molecular dynamics simulation package ‘GROMACS’

\* Corresponding author.

E-mail addresses: [lpusztai@szfki.hu](mailto:lpusztai@szfki.hu), [laszlo.pusztai@gmail.com](mailto:laszlo.pusztai@gmail.com) (L. Pusztai).

[11]) and shorter (1.25 fs) time step, as in this calculation we tried to reproduce the simulations performed by Svishchev and Kusalik [12] to make the comparison with the results of these authors more accurate.

For a comparison with a different potential model, the GROMACS-adapted [13] version of the SWM4-DP potential was chosen (calculation name: 'SWM4-DP') which, in terms of number of particles and temperature, was identical to the H3 calculation. In our version of the SWM4-DP model the Drude particle was massless, no second thermostat for the oxygen–Drude pair was introduced and a time step of  $dt = 2$  fs was applied (see Table 1 for the summary of systems used). This simulation, just as in the original work of Lamoureux, was carried out in the NPT ensemble at a pressure of 1 bar.

All the molecular dynamics simulations were carried out by the GROMACS simulation package [11], using the leap-frog algorithm for integrating Newton's equations of motion with a time step  $dt = 2$  fs, if not stated otherwise. For checking possible effects of this choice, an additional calculation with a time step of  $dt = 1$  fs and trajectory length of 7600 ps was performed for the H1 system. Deviations, in terms of the dielectric constant, between the two time step values were observable only up to  $\sim 2000$  ps; beyond this point convergence was identical (not shown). Based on this extra simulation, there is no difference whether 1 or 2 fs time steps are used concerning the calculation of the dielectric constant, provided that a sufficiently long trajectory is sampled. For avoiding prohibitively long calculations, the  $dt = 2$  fs time step was adopted.

The main simulation parameters of the models are listed in Table 1. All simulations used the 6–12 LJ potential with a cutoff of 0.9 nm. The rigid molecular geometry was enforced with the SETTLE [14] algorithm. Calculations H1, H2, H3, H4 and SWM4-DP were carried out with the same parameters (except for the differences explicitly given in Table 1) using particle-mesh Ewald [15] summation for the long range Coulomb-forces with interpolation order 4 and a grid size of 0.12 nm. The relative strength of the electrostatic interaction at the cutoff was  $\sim 10^{-5}$ , providing an accuracy of  $5 \times 10^{-3}$  which is still higher than the accuracy of the Lennard–Jones calculations. LJ parameters for SPC/E oxygen were  $\sigma = 0.316557$  nm,  $\epsilon = 0.650194$  kJ/mol, and  $\sigma = 0.31803$  nm and  $\epsilon = 0.861141024$  kJ/mol for the SWM4-DP model. The temperature (and in case of the SWM4-DP model, the pressure, as well) was held constant with the help of a Berendsen thermostat [16] with temperature coupling time constant  $\tau_T = 0.1$  ps (pressure coupling time constant  $\tau_P = 1$  ps). For checking the possible influence of the thermostat, the entire calculation on model H1 has been repeated by using the Nose–Hoover thermostat [17]; this calculation will be denoted as H1-NH.

The equilibration stage, the first 400 ps of the simulation, was discarded; data collection was carried out for the following 6–8.2 ns, as it is given in Table 1. Coordinates were collected at each time step for each system, apart from the largest H4 model,

where every 10-th configuration was collected (The size of the configuration file of the mid-size 2000 molecule systems was around 300 Gbytes, which made the storage and handling of the file difficult).

### 3. Static dielectric constant

During the calculation of the static dielectric constant (relative static permittivity),  $\epsilon$ ,  $\epsilon_{\text{RF}} = \infty$  was assumed.  $\epsilon$  was calculated for the SPC/E model according to the formula

$$\epsilon = 1 + \frac{\langle M^2 \rangle - \langle M \rangle^2}{3\epsilon_0 V k_B T}; \quad (1)$$

and for the polarizable SWM4-DP model, similarly as in Lamoureux [6], using

$$\epsilon = \epsilon_\infty + \frac{\langle M^2 \rangle - \langle M \rangle^2}{3\epsilon_0 V k_B T}, \quad \frac{\epsilon_\infty - 1}{\epsilon_\infty + 2} = \frac{4\pi\alpha}{3\langle v \rangle}; \quad (2)$$

where  $M$  is the total dipole moment,  $\epsilon_0$  is the vacuum permittivity,  $V$  is the volume of the system,  $k_B$  is the Boltzmann-constant,  $T$  is the temperature,  $\langle v \rangle$  is the average molecular volume and  $\alpha$  is the polarizability. For the polarizability for the SWM4-DP model the value of  $\alpha = 1.04252 \text{ \AA}^3$  from Lamoureux et al. [18] was used. The average molecular volume was  $\langle v \rangle = 30.49478 \text{ \AA}^3$ , yielding  $\epsilon_\infty = 1.5$ . All calculations applied the *g\_dipoles* software provided by the GROMACS package [11].

Note that the exact equation by which the dielectric constant was calculated is not really important here (even though a large volume of the literature concerns this issue, see e.g. [19,20]). What we wish to scrutinize is the convergence properties of  $\langle M^2 \rangle - \langle M \rangle^2$  with time (see Eqs. (1) and (2)). It is clear that the net (or 'total') dipole moment for the simulation box,  $\langle M \rangle$ , is a quantity of utmost importance here and therefore, its behavior has also been monitored for the H1 and H2 models. In both cases,  $\langle M \rangle$  had relatively large values at the beginning of the simulation, of the order of 100 D; these values have continuously decreased to just 2–5 D by the end of the simulations. These values represent a rather good approach to the ideal limit of 'zero net dipole' moment.

For the largest H4 system it was not possible to collect every consecutive configuration for 8 ns (due to the exceptionally large storage space requirement); for this reason, the effect of sampling only every  $x$ -th configuration was also studied. Results for each model are given in Table 2. There is no significant change if the dielectric constant is calculated based on every  $x = 1$ –1000 configuration; on the other hand,  $x = 10000$  results in significantly

**Table 2**

The static dielectric constant calculated from every  $x$ th configuration of the trajectory. 'ε final' is the final value,  $\bar{\epsilon}$  is the average and  $\sigma$  is the standard deviation calculated for the properly converged, last 3000 ps part of the  $x = 1$  trajectory. There was too calculations performed for the H1 system, using a bit different parts (0–8000 or 200–8200 ps) of the trajectory.

| Model            | ε Final |          |           |            |             | $\bar{\epsilon}$<br>(σ) |
|------------------|---------|----------|-----------|------------|-------------|-------------------------|
|                  | $x = 1$ | $x = 10$ | $x = 100$ | $x = 1000$ | $x = 10000$ | $x = 1^a$               |
| H1 (0–8000 ps)   | 69.8    | –        | 69.8      | 69.7       | 68.1        | 70.6 (0.40)             |
| H1 (200–8200 ps) | 69.5    | –        | 69.5      | 69.5       | 68.0        | 70.3 (0.48)             |
| H2               | 65.6    | –        | 65.7      | 66.5       | 66.5        | 65.4 (0.44)             |
| H3               | 77.9    | –        | 77.9      | 77.9       | 79.4        | 78.2 (0.40)             |
| H4               | –       | 72.4     | 72.4      | 72.3       | 69.7        | 72.4 (0.66)             |
| H1-O             | 67.1    | –        | 67.1      | 66.9       | 66.0        | 67.9 (0.71)             |
| H3-T             | 72.3    | –        | 72.3      | 72.5       | 68.4        | 71.5 (0.59)             |
| SWM4-DP          | 78.1    | –        | 78.2      | 77.4       | 78.8        | 78.9 (0.55)             |
| Exp (293 K)      |         |          |           |            |             | 80.4 [23]               |
| Exp (298 K)      |         |          |           |            |             | 78.4 [24]               |

<sup>a</sup> The average for the H4 system is calculated from the  $x = 10$  curve.

**Table 1**

Basic parameters of the simulation models: name, chemical composition, number of molecules ( $N$ ), temperature, potential model, time step ( $dt$ ), data collection length for the coordinates.

| Name    | System           | $N$   | $T$<br>(K) | Potential | $dt$<br>(fs) | Length for<br>coordinates<br>(ns) |
|---------|------------------|-------|------------|-----------|--------------|-----------------------------------|
| H1      | H <sub>2</sub> O | 216   | 293        | SPC/E     | 2            | 8.2                               |
| H2      | H <sub>2</sub> O | 518   | 293        | SPC/E     | 2            | 8                                 |
| H3      | H <sub>2</sub> O | 2000  | 293        | SPC/E     | 2            | 8                                 |
| H4      | H <sub>2</sub> O | 16000 | 293        | SPC/E     | 2            | 8                                 |
| H1-O    | H <sub>2</sub> O | 216   | 298        | SPC/E     | 1.25         | 6                                 |
| H3-T    | H <sub>2</sub> O | 2000  | 298        | SPC/E     | 2            | 8                                 |
| SWM4-DP | H <sub>2</sub> O | 2000  | 293        | SWM4-DP   | 2            | 8                                 |

different, sometimes lower, sometimes higher values. We can conclude that it is possible to use every 10–1000th, or, to be on the safe side, every 10–100th configuration at least for this type of system, so that storage capacity and computation time can be optimized.

Convergence of the dielectric constant for the H3 model is given in Figure 1a. The curves for the  $x = 1$  and 100 are identical for the entire period. The  $x = 1000$  curve joins them around  $t = 1.4$  ns, while the curve associated with the  $x = 10\,000$ -th configuration sampling is distinctly different even in terms of the asymptotic value. This behavior is the same for each model in this work, which suggests that this is a general pattern.

Considering the convergence of the static dielectric constant it is clear that proper convergence can only be achieved after 3000–5000 ps. Usually much shorter time ranges have been used for the calculation of the static dielectric constant [6,12,21,22] which makes earlier results somewhat questionable. In Table 2 the average and the standard deviation calculated for the properly converged last 3000 ps of the trajectory are shown for  $x = 1$  curves ( $x = 100$  for H4).

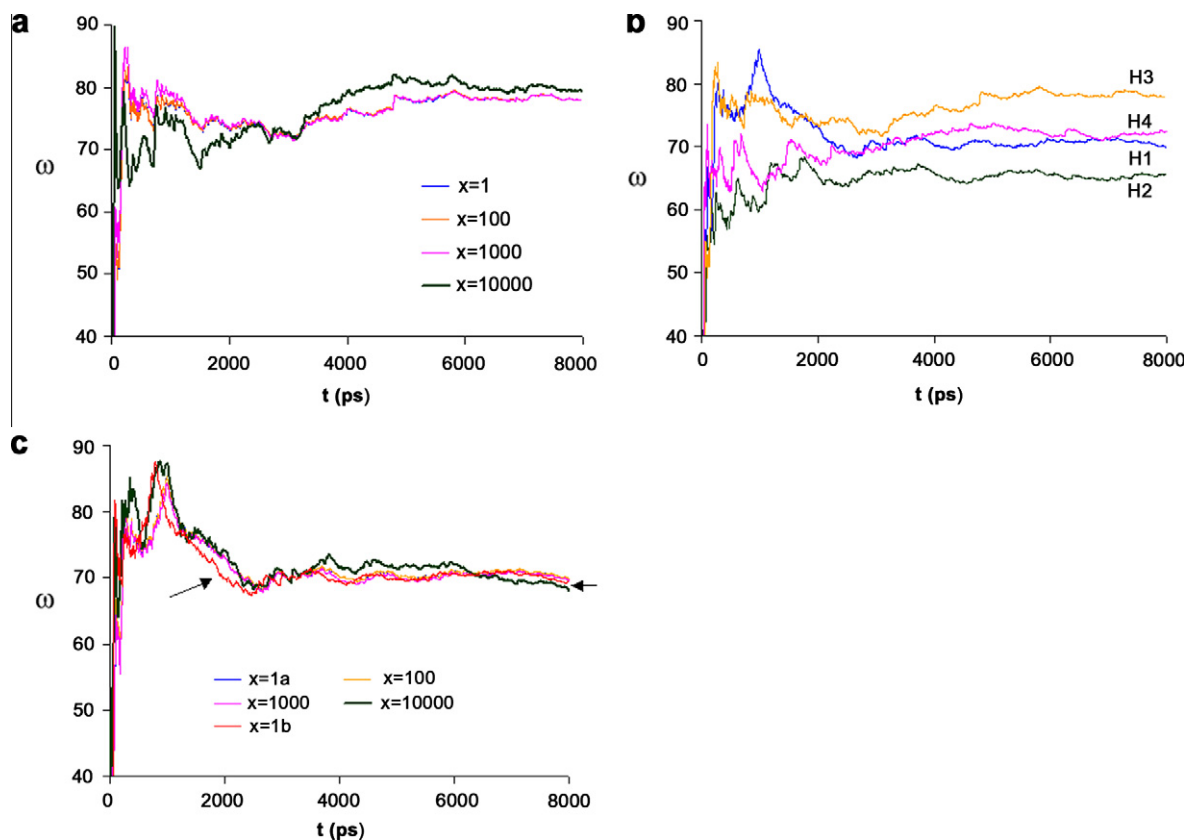
Now let us turn to the system size dependence of the dielectric constant, as revealed here on the examples of H1, H2, H3 and H4 models. Unfortunately, there is quite a large difference (68.3–78.2) between the models, and no monotonous system size dependence can be established (Table 2, Figure 1b). This is similar to Morrow's findings [9], based on theoretical calculations for dipole lattices in case of reaction field, where the convergence was substantially oscillating with system size. Deviations between the different model sizes are much larger than the fluctuations of the static dielectric constant for any one model over the last 3000 ps

of their trajectories, so they cannot be explained by improper convergence. To see how big a change in terms of the value of the dielectric constant can be found due to different starting points used in the calculation within the same model,  $\epsilon$  was calculated with  $x = 1$  from 200 to 8200 ps relative to the beginning of the data collection for the H1 model (see Figure 1c). Although there is a marked difference at the beginning of the simulations up to 4000 ps between this and the previous (0–8000 ps) calculations, later the dielectric constant would converge virtually to the same value as before (see Table 2).

Comparing the 2000 molecule 293 K H3 and 298 K H3-T models the correct behavior, the decrease of the dielectric constant with increasing temperature, can be observed. The same can be seen for the 293 K H1 and 298 K H1-O systems, although here not only the temperature but also the simulation parameters differ slightly.

If we compare our  $\bar{\epsilon}$  with the value given for SPC/E water using a 500 ps trajectory of a 256 molecule system at 298 K by Svishchev [12], the static dielectric constant given by them is  $71.0 \pm 6.1$  for  $\text{H}_2\text{O}$  and  $68.5 \pm 6.6$  for  $\text{D}_2\text{O}$ . The  $\epsilon$  value for our H1-O model is  $67.9 \pm 1.5$ , which is in agreement with that. It has to be noted that due to the much longer trajectory length, our standard deviation is much smaller. The values for the  $\text{D}_2\text{O}$  models cannot be directly compared since the temperature in Svishchev's calculations was 298 K while in ours it was 293 K. Still, it has to be noted that no difference is expected due to the isotopic substitution, so the—slightly different—values found for the otherwise identical H1 ( $70.6 \pm 0.4$ ) and D1 ( $71.5 \pm 0.53$ ) models can only be attributed to numerical uncertainties.

It is worth mentioning that changing the thermostat (from the Berendsen type to the Nose–Hoover type, see model H1-NH) does



**Figure 1.** Convergence of the static dielectric constant for (a) the H3 model; (b) for models H1, H2, H3, H4 ( $x = 100$  for all four systems); (c) H1 model. Every  $x$ th configuration was used for the calculation of the running averages;  $x$  is given on the graph if not stated explicitly here. In case of (c) a second calculation range for the H1 system with  $x = 1$  from 200 to 8200 ps is also shown, which is indicated by two arrows for clarity.

change the actual converged value of  $\varepsilon$  to about 74, which is a roughly 5% increase. That is, when mentioning the dielectric constant of a given water model the thermostat must also be mentioned. Note, however, that the convergence properties are remarkably similar: the Nose–Hoover thermostat needs even longer trajectories, up to about 7 ns, for a proper convergence.

For the SWM4-DP model the original value of Lamoureux at 298 K was  $79 \pm 5$  [6], while our 293 K simulation yielded  $78.9 \pm 0.55$ , which also are in good agreement.

#### 4. Conclusions

- (1) It was demonstrated that shorter than about 5(–6) ns trajectories are not applicable for providing a good estimate of the dielectric constant for any system size considered here; any value resulting from shorter calculations must be handled with care. It should be noted that, strictly speaking, calculation of the dielectric constant does not necessitate the presence of time dependence (dynamics); a statement that would be equivalent to the above could be that a volume of the configuration space that is comparable to the section sampled during about 5–6 ns in a molecular dynamics simulation should be used.
- (2) The static dielectric constant changes strongly with the system size, too; however, the monotonous increase with the system size predicted by Morrow [9] could not be verified. Even long trajectory lengths cannot ensure an infinitely accurate estimation of the dielectric constant, due to the observed system size dependence: the difference between extremes is about 10%.
- (3) The method chosen for thermostating the simulated system changes the converged value of the dielectric constant (in our case, by about 5%), but not the need for long (at least 6 ns for the Nose–Hoover thermostat) simulated trajectories.

#### Acknowledgment

LP acknowledges financial help from the Hungarian Basic Research Fund (OTKA), via grant No. K083529.

#### References

- [1] H.J.C. Berendsen, J.R. Grigera, T.P. Straatsma, *J. Phys. Chem.* 91 (1987) 6269.
- [2] P.G. Kusalik, I.M. Svishchev, *Science* 265 (1994) 1219.
- [3] I.M. Svishchev, P.G. Kusalik, *J. Chem. Phys.* 99 (1993) 3049.
- [4] I.M. Svishchev, P.G. Kusalik, *Chem. Phys. Lett.* 215 (1993) 596.
- [5] C. Vega, J.L.F. Abascal, M.M. Conde, J.L. Aragones, *Faraday Discuss.* 141 (2009) 251.
- [6] G. Lamoureux, A.D. MacKerell Jr., B. Roux, *J. Chem. Phys.* 119 (2003) 5185.
- [7] V. Mile, L. Pusztai, H. Dominguez, O. Pizio, *J. Phys. Chem. B* 113 (2009) 10760.
- [8] V. Mile, Personal Communication, Research Institute for Solid State Physics and Optics, Hungarian Academy of Sciences, Budapest, Hungary, 2010.
- [9] T.J. Morrow, E.R. Smith, *J. Stat. Phys.* 61 (1990) 187.
- [10] J. Casulleras, E. Guardia, *Mol. Sim.* 8 (1992) 273.
- [11] <http://www.gromacs.org>.
- [12] I.M. Svishchev, P.G. Kusalik, *J. Phys. Chem.* 98 (1994) 728.
- [13] B. Hess, C. Kutzner, D. van der Spoel, E. Lindahl, *J. Chem. Theory Comput.* 4 (2008) 435.
- [14] S. Miyamoto, P.A. Kollman, *J. Comp. Chem.* 13 (1992) 952.
- [15] U. Essman, L. Perela, M.L. Berkowitz, T. Darden, H. Lee, K.G. Pedersen, *J. Chem. Phys.* 103 (1995) 8577.
- [16] H.J.C. Berendsen, J.P.M. Postma, W.F. van Gunsteren, A. DiNola, J.R. Haak, *J. Chem. Phys.* 81 (1984) 3684.
- [17] S. Nose, *Prog. Theor. Phys. Suppl.* 103 (1984) 1; W.G. Hoover, H.A. Posch, B. Holian, M.J. Gillan, M. Mareschal, C. Massobrio, *Mol. Simul.* 1 (1987) 79.
- [18] G. Lamoureux, E. Harder, I.V. Vorobyov, B. Roux, A.D. MacKerell Jr., *Chem. Phys. Lett.* 418 (2006) 245.
- [19] M. Neumann, *Mol. Phys.* 50 (1983) 841.
- [20] S.W. de Leeuw, J.W. Perram, E.R. Smith, *Proc. R.S. London A* 373 (1980) 27.
- [21] P.J. van Maaren, D. van der Spoel, *J. Phys. Chem.* 105 (2001) 2618.
- [22] A. Wallqvist, B.J. Berne, *J. Phys. Chem.* 97 (1993) 13841.
- [23] A.M. Bottreau, J.M. Moreau, J.M. Laurant, C. Marzat, *J. Chem. Phys.* 62 (1975) 360.
- [24] J.N. Murrell, A.D. Jenkins, *Properties of Liquids and Solutions*, second ed., John Wiley & Sons, Chichester, England, 1994.